POSTGRADUATE DIPLOMA IN ICT

Capstone Project — Implementation & Testing/Evaluation Submission

Project Title	*[Enter full project title]*
Student Full Name	*[Enter full name]*
Student Number	*[Enter student number]*
Supervisor Name	*[Enter supervisor's full name and title]*
Stream Category	*[Select ONE: Application Development | Computer Networks | AI and Data Science | Web Technology | Information Systems]*
Academic Year	*[e.g. 2025 / 2026]*
Submission Date	*[DD Month YYYY]*
Repository / Code Location	*[Git URL, or note that a zipped archive is attached]*

Declaration

I declare that the implementation, testing, and evaluation described in this submission is my own original work and has been prepared in accordance with the academic integrity policies of the institution.

Student Signature	_____
Date	_____

1. Purpose of This Document

This document accompanies your implemented system and its source code/repository. It is assessed against the Implementation & Testing/Evaluation Rubric. Where a section refers to the codebase, ensure the referenced code is present and accessible in the repository/archive submitted alongside this document.

2. Implementation Summary [35 Marks]

This section carries 35 marks: Code Structure, Adherence to Design & Best Practices (18) and Functionality Completeness Against Requirements (17).

2.1 Adherence to Approved Design

Explain how the implemented architecture and module structure follows the approved Design Phase HLD/DDD. Explicitly document and justify any deviations from the approved design.

[Describe how the implementation follows the Design Phase artefacts, and justify any deviations]

2.2 Requirements Implementation Status

List every functional requirement from the Analysis/Design phases and its current implementation status. Use the same requirement IDs used in your Design Phase traceability matrix.

Requirement ID & Description	Status	Notes / Evidence Location
[REQ-001: Provide requirement description]	*[Implemented / Partially Implemented / Not Implemented]*	*[File/module or demo reference]*
[REQ-002]	*[Status]*	*[Notes]*

[REQ-003]	*[Status]*	*[Notes]*
[Add rows as required]		

2.3 Code Structure & Best Practices

Describe how the codebase is organised (modules/packages/classes), naming and style conventions followed, and any best practices applied (e.g. version control workflow, error handling, dependency management).

[Describe code structure, conventions, and best practices applied]

3. Testing Strategy & Execution [30 Marks]

This section carries 30 marks: Test Plan Design & Coverage (15) and Test Execution, Results & Defect Management (15).

3.1 Test Plan

Define your test cases, each traced to a specific functional or non-functional requirement. Include unit, integration, and (where applicable) system/user acceptance test levels.

Test Case ID	Requirement(s) Traced	Test Level	Test Description	Expected Result
[TC-001]	*[REQ-001]*	*[Unit/Integration/UAT]*	*[What is being tested]*	*[Expected outcome]*
[TC-002]	*[REQ-...]*	*[Level]*	*[Description]*	*[Expected outcome]*
[Add rows as required]				

3.2 Test Execution Results

Report the actual outcome of each test case executed. Attach logs, screenshots, or output data as evidence in the Appendices.

Test Case ID	Actual Result	Pass / Fail	Evidence Reference (Appendix)
[TC-001]	*[What actually happened]*	*[Pass / Fail]*	*[Appendix A, Fig. ...]*
[TC-002]	*[Actual result]*	*[Pass / Fail]*	*[Appendix reference]*
[Add rows as required]			

3.3 Defect Log & Resolution

For every failed test or defect discovered, record it here with its priority and current resolution status.

Defect ID	Description	Priority (H/M/L)	Status / Resolution
[DEF-001]	*[Describe the defect]*	*[H/M/L]*	*[Fixed / Deferred (with justification) / Open]*
[Add rows as required]			

4. Evaluation & Results Analysis [25 Marks]

This section carries 25 marks: Presentation & Analysis of Evaluation Results (13) and Critical Evaluation Against Project Objectives (12).

4.1 Evaluation Results

Present the evidence used to judge system quality — performance metrics, benchmark comparisons, user feedback, or other evidence appropriate to your project type. Use tables/charts and reference them from the Appendices where large.

[Present and interpret your evaluation results (metrics, comparisons, feedback, etc.)]

4.2 Critical Evaluation Against Project Objectives

Revisit each objective stated in your Design/Analysis Phase documents. For each, state honestly whether it was met, partially met, or not met, and why.

Project Objective	Met / Partially Met / Not Met	Explanation
[Objective 1]	*[Status]*	*[Explanation, including any limitations]*
[Objective 2]	*[Status]*	*[Explanation]*
[Add rows as required]		

5. Implementation & Testing Documentation [10 Marks]

This section carries 10 marks: Technical Documentation Quality (10).

5.1 Setup & Installation Instructions

Provide instructions sufficient for another developer to install dependencies, configure, and run the system from the submitted repository/archive.

[Provide setup, dependency, and run instructions]

5.2 Key Technical Decisions

Summarise any significant technical decisions made during implementation that were not already covered in the Design Phase (e.g. library substitutions, workarounds for constraints discovered during development).

[Summarise key implementation-stage technical decisions]

6. Self-Reflection

This reflects Section E of the rubric (holistic — no independent marks, but informs overall mark quality): Code Quality, Security & Maintainability; Academic Writing, Referencing & Presentation.

[Reflect on the security and maintainability of your implementation, and any academic sources referenced]

7. References

List all sources cited in this document using the institution's prescribed referencing style (typically Harvard or APA 7th edition).

[List references in the prescribed format]

8. Appendices

Appendix A — Test Execution Evidence

[Insert test logs, screenshots, or output data referenced in Section 3.2]

Appendix B — Representative Code Snippets

Include representative code snippets illustrating key functionality. All code must be formatted in a monospaced font with line numbers. Do not paste the entire codebase — reference the repository for full source.

[Insert representative code snippets]

9. Self-Assessment Checklist — Rubric Alignment

Complete this checklist before submitting. Each criterion maps directly to the Implementation & Testing/Evaluation Rubric.

Section Reference	Assessment Criterion	Max Marks	Self-Score
Section 2	Implementation Summary	35	*[/ 35]*
Section 3	Testing Strategy & Execution	30	*[/ 30]*
Section 4	Evaluation & Results Analysis	25	*[/ 25]*
Section 5	Implementation & Testing Documentation	10	*[/ 10]*
	TOTAL	**100**	*[/ 100]*

Criterion	Completed? (Yes / Partial / No)
Every requirement from the Design Phase traceability matrix has a recorded implementation status	*[Yes / Partial / No]*
Test cases are explicitly traced to specific requirements, across appropriate test levels	*[Yes / Partial / No]*
Test execution results are reported honestly, including any failures, with evidence attached	*[Yes / Partial / No]*
A defect log is included with priority and resolution status for every issue found	*[Yes / Partial / No]*
Evaluation results are presented with appropriate evidence (metrics/charts/feedback)	*[Yes / Partial / No]*
Every project objective is critically assessed as met, partially met, or not met, with explanation	*[Yes / Partial / No]*
Setup/installation instructions are complete enough for another developer to run the system	*[Yes / Partial / No]*
Document has been proofread and uses formal academic language throughout	*[Yes / Partial / No]*

— End of Implementation & Testing/Evaluation Submission Template —